

DS 203: Mathematical Foundations of Data Science I

Course Code: DS 203

Course Name: Mathematical Foundations of Data Science I

L-T-P-C: 3-0-0-3

Prerequisites: IC 152: Data Science I

Intended for: UG

Distribution: Discipline core for B. Tech. Data Science and Engineering, Not allowed to B.Tech. CSE, Free elective for other B. Tech. disciplines.

Semester: Even/Odd.

Approval: 29th BoA (B.Tech DSE curriculum approved by the 20th Senate)

Objective

During the last two decades, there has been a major paradigm shift in processing, communication, and storage of information from predominantly analog domain to digital domain for the reasons such as ease of implementation, better efficiency, greater robustness against noise, and enhanced performance and security. This shift has not only resulted in generation of ever increasing volumes of digital data but also an acute need to efficiently process, store, and communicate it. To address this need the focus of DSE curriculum must shift from introducing the traditional elementary discrete structures to represent and process the data to more abstract structures provided by fields such as graph theory. This will enable the students to better understand and appreciate the current developments at the frontiers of DS, both in theory and application, and prepare them to contribute to further advancement of such frontiers.

On completion of this course, students should be able to demonstrate their understanding of and apply methods of discrete mathematics in DS to subsequent courses in algorithm design and analysis, automata theory and computability, knowledge representation, information systems, information security. In particular, students should be able to

- use logical notation to define fundamental mathematical concepts such as sets, relations, functions and various algebraic structures, reason mathematically using such structures, and evaluate arguments that use such structures.
- model and analyse a computation process and construct elementary proofs based on such structures

Course Modules

- **Fundamental structures:**
 - Functions: Surjections, injections, inverses, composition. (4 lectures)
 - Relations: Reflexivity, symmetry, transitivity, equivalence relations. (2 lectures)
 - Sets: Venn diagrams, complements, Cartesian products, power sets, finite and infinite sets, introduction to lattices. (5 lectures)
 - Abstract orders: Quasi-order, partial order, well-order, (Advanced, optional topics: Zorn's lemma, Koenig's theorem.) (4 lectures)
- **Combinatorics:** Counting arguments/techniques; pigeonhole principle; cardinality and countability, the inclusion-exclusion principle, recurrence relations, generating functions. (6 lectures)

- **Basics of graph theory:** Graph as a discrete structure, graph coloring and connectivity, traversal problems, and spanning trees. (3 lectures)
- **Introduction to graph mining:** Large-scale connected components, similarity and centrality rankings, clustering. (3 lectures)
- **Logic:** Propositional and predicate logic: syntax, semantics, soundness, completeness, unification, inferencing, resolution principle, proof system. (3 lectures)
- **Proof techniques:** Negation, contradiction, contraposition, mathematical induction) and the structure of formal proofs; efficiency of proof-systems. (4 lectures)
- **Semantics:** Introduction to operational semantics, structural operational semantics, Hoare logic. (3 lectures)
- **State machines:** Introduction, minimization, grammars, languages. (5 lectures)

Text Books

1. Lehman, E., Leighton, F. T., and Meyer A. R., *Mathematics for Computer Science*, 2013.

Reference Books

1. Graham, R. L., Knuth, D. E., and Patashnik O., *Concrete Mathematics*, Pearson, 1994.
2. Aho, Alfred V., and Jeffrey D. Ullman. *Foundations of computer science*. Computer Science Press, Inc., 1992.
3. Tucker, A., *Applied Combinatorics*, Wiley, 2012.
4. Liu, A. and Mohapatra, D. P., *Elements of Discrete Mathematics*, Tata-McGraw Hill, 2008.
5. Koshy T., *Discrete Mathematics with Applications*, Academic Press, 2003.
6. Hein J., *Discrete Structures, Logic, and Computability*, Jones and Barlett, 2009.